Unit 5: Data Analytics

Complete exercise solutions — MCQs, Short Questions, Long Questions

EXERCISE SOLUTIONS — Unit 5: Data Analytics

Multiple Choice Questions

1. An example of a basic statistical model is:

• a) Linear Regression

• b) Neural Networks

• c) Decision Trees

• d) Support Vector Machines

Answer: a) Linear Regression

2. The activity involved in experimental design in data science is:

• a) Creating visualizations

• b) Collecting and analyzing data systematically

• c) Writing code for machine learning

• d) Building databases

Answer: b) Collecting and analyzing data systematically

3. A commonly used tool for creating data visualizations is:

• a) MS Excel

• b) Python (Matplotlib)

• c) Tableau

• d) All of the above

Answer: d) All of the above

4. The meaning of the slope in a linear regression model is:

• a) The intercept of the model

• b) The change in the dependent variable for a unit change in the independent variable

• c) The error term

• d) The mean of the data

Answer: b) The change in the dependent variable for a unit change in the independent variable

5. An example of a real-world application of statistical models is:

• a) Predicting house prices

• b) Creating social media posts

• c) Designing websites

studysapiens.com

• d) Writing essays

Answer: a) Predicting house prices

6. Option NOT considered a benefit of data visualization is:

• a) Identifying trends and patterns

• b) Communicating insights effectively

• c) Making data more complex

• d) Summarizing large datasets

Answer: c) Making data more complex

7. A primary goal of K-Means Clustering is:

• a) To classify data into predefined categories

• b) To group data into clusters based on similarity

• c) To predict continuous outcomes

• d) To reduce the dimensionality of data

Answer: b) To group data into clusters based on similarity

8. The meaning of 'K' in K-Means Clustering is:

• a) Number of features in the dataset

• b) Number of clusters to be formed

• c) Number of iterations required for convergence

• d) Number of data points in the dataset

Answer: b) Number of clusters to be formed

Short Questions

1. What is the importance of building statistical models in real-world applications?

Statistical models help us understand data, make predictions, and support decision-making. Their importance: (1) they summarise large datasets into simple, understandable forms; (2) they predict future outcomes, such as house prices or sales; (3) they reveal patterns and relationships that are not obvious in raw data; (4) they support evidence-based decisions in business, healthcare, and government. Without models, decisions would rely on guesswork.

2. Name one basic statistical model used for predicting outcomes and explain its purpose.

Linear Regression is a basic statistical model used for predicting a continuous outcome. It models the relationship between a dependent variable (Y) and one or more independent variables (X) using the equation $Y = \beta0 + \beta1X$, where $\beta0$ is the intercept and $\beta1$ is the slope. Its purpose is to predict the value of Y for given values of X. Example: predicting daily earnings (Y) from the number of customers (X) at a fruit stall — with slope 40 and intercept 100, Earnings = 100 + 40 × Customers.

3. List two types of data visualizations and describe when you would use each.

(1) Bar Charts — used to compare values across categories. Example: comparing test scores of Class A and Class B. (2) Line Charts — used to show trends over time. Example: tracking monthly sales or temperature changes throughout a year. Other types include pie charts (for proportions), scatter plots (for relationships between two variables), and histograms (for distributions). The choice depends on what the data represents and what insight you want to show.

4. How does visualizing data help in understanding descriptive statistics?

Visualizing data makes descriptive statistics easier to understand because: (1) patterns, trends, and outliers become visible at a glance; (2) the shape of the distribution (symmetric, skewed, clustered) can be seen, which helps interpret mean, median, and mode; (3) comparisons between groups become clear; (4) anomalies and errors in data can be spotted. For example, a histogram of test scores immediately shows whether most students scored high or low, which the mean alone might hide.

Long Questions

1. Explain the role and importance of statistical models in solving real-world problems.

Statistical models play a central role in solving real-world problems because they: (1) summarise large datasets into meaningful measures like mean, median, and standard deviation; (2) identify relationships between variables, such as how advertising affects sales; (3) predict future outcomes, enabling planning and forecasting; (4) support decisions in fields like healthcare (predicting disease risk), finance (credit scoring), agriculture (crop yield prediction), and education (student performance). Models turn raw data into actionable knowledge, replacing intuition with evidence.

2. Describe the steps involved in building a basic statistical model (e.g., linear regression). Include details on data collection, model training, and evaluation.

Steps: (1) Data Collection — gather relevant data from surveys, observations, or experiments. (2) Data Cleaning — handle missing values (imputation, flagging, or removal) and remove errors. (3) Data Exploration — visualise the data and compute descriptive statistics. (4) Model Training — fit the model to the data; for linear regression, find the slope and intercept that minimise prediction errors. (5) Model Evaluation — test the model on new data using metrics like R-squared or RMSE to check accuracy. (6) Deployment — use the trained model to make predictions. Each step is essential for a reliable model.

StudySapiens.com

3. Discuss the types of data visualizations and their uses.

Common types: (1) Bar charts — compare values across categories (e.g., sales by region). (2) Line charts — show trends over time (e.g., monthly revenue). (3) Pie charts — show proportions of a whole (e.g., market share). (4) Scatter plots — show relationships between two variables (e.g., study hours vs. marks). (5) Histograms — show the distribution of a single variable (e.g., test score distribution). (6) Box plots — summarise spread and outliers. Each type suits a different kind of data and question. Tools like Excel, Tableau, and Matplotlib create these visualizations.

4. Explain data collection methods.

Data collection methods include: (1) Surveys — questionnaires given to a sample of people. Example: a grocery store surveying customers on product preferences. (2) Observations — watching and recording behaviour. Example: a restaurant tracking which tables customers choose. (3) Experiments — controlled tests to measure cause and effect. Example: a teacher testing whether printed notes improve exam performance. (4) Secondary data — using existing records from organisations. The choice depends on the research question, budget, and time available. Good collection ensures the data is accurate and representative.

5. Discuss the concept of measures of central tendency with examples.

Measures of central tendency describe the 'centre' or typical value of a dataset. There are three: (1) Mean — the average of all values (sum divided by count). Example: scores 50, 60, 70, 80, 90 have a mean of 70. (2) Median — the middle value when data is arranged in order. Example: the same scores have a median of 70. (3) Mode — the value that appears most often. Example: in 50, 60, 60, 70, 80, the mode is 60. The mean is affected by outliers, while the median is more robust. The mode is useful for categorical data. Together they give a fuller picture of the data.

studysapiens.com